

Comparative Effectiveness of Metformin, Sulfonylureas, and DPP-4 Inhibitors as Second-Line Agents for Type 2 Diabetes: A Systematic Review and Network Meta-Analysis

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ABSTRACT

BACKGROUND Multiple oral agents are available as second-line therapies for Type 2 Diabetes when metformin monotherapy is insufficient. Direct comparative evidence is limited. **METHODS** We conducted a systematic review and network meta-analysis of randomized controlled trials comparing second-line therapies added to metformin. Outcomes included HbA1c reduction, weight change, and hypoglycemia risk. **RESULTS** GLP-1 Agonists and SGLT2 Inhibitors provided superior HbA1c reduction (-0.9% to -1.2%) compared to DPP-4 Inhibitors (-0.6%) and Sulfonylureas (-0.8%). GLP-1 Agonists lead to significant weight loss (-2.8 kg), while Sulfonylureas caused weight gain (+2.3 kg). SGLT2 Inhibitors were weight-neutral. **CONCLUSIONS** GLP-1 Agonists and SGLT2 Inhibitors offer superior glycemic control and favorable weight profiles compared to traditional second-line agents for Type 2 Diabetes.

INTRODUCTION

Type 2 Diabetes Mellitus affects over 500 million individuals globally. Metformin remains the first-line pharmacotherapy due to its efficacy, safety profile, and low cost. However, progressive beta-cell dysfunction necessitates intensification of therapy in most patients within 3-5 years.

Second-line options include Sulfonylureas, DPP-4 Inhibitors, GLP-1 Receptor Agonists, SGLT2 Inhibitors, and Insulin Therapy. Clinical guidelines provide multiple options but limited head-to-head comparative data to guide selection.

METHODS

We searched MEDLINE, Embase, and CENTRAL for randomized trials published through December 2023. Eligible studies compared second-line agents in adults with Type 2 Diabetes inadequately controlled on metformin (HbA1c $\geq 7.0\%$).

Network meta-analysis was conducted using a Bayesian framework. Primary outcome was change in HbA1c at 24-52 weeks. Secondary outcomes included body weight, hypoglycemia, and treatment discontinuation.

RESULTS

A total of 127 trials (n=58,453 participants) were included.

HbA1c Reduction: - GLP-1 Agonists (Semaglutide, Liraglutide): -1.2% (95% CrI: -1.4 to -1.0) - SGLT2 Inhibitors (Dapagliflozin, Empagliflozin): -0.9% (95% CrI: -1.1 to -0.7) - Sulfonylureas (Glipizide, Gliclazide): -0.8% (95% CrI: -1.0 to -0.6) - DPP-4 Inhibitors (Sitagliptin, Linagliptin):

-0.6% (95% CrI: -0.8 to -0.4) - Insulin Therapy (Basal Insulin): -1.1% (95% CrI: -1.3 to -0.9)

Weight Change: - GLP-1 Agonists caused the greatest weight loss (-2.8 kg) - SGLT2 Inhibitors provided moderate weight loss (-1.8 kg) - DPP-4 Inhibitors were weight-neutral - Sulfonylureas and Insulin Therapy both caused weight gain (+2.3 kg and +3.1 kg respectively)

Hypoglycemia: - Sulfonylureas had the highest risk (RR 3.8 vs placebo) - Insulin Therapy also elevated risk (RR 2.9) - GLP-1 Agonists, SGLT2 Inhibitors, and DPP-4 Inhibitors had minimal hypoglycemia risk

DISCUSSION

This network meta-analysis demonstrates that GLP-1 Receptor Agonists provide the most favorable overall profile for second-line Type 2 Diabetes therapy, with superior HbA1c reduction, substantial weight loss, and minimal hypoglycemia risk. SGLT2 Inhibitors offer similar benefits with added cardiovascular and renal protection demonstrated in outcomes trials.

Traditional agents like Sulfonylureas remain widely used due to low cost and familiarity, but their weight gain and hypoglycemia risk are clinically significant drawbacks. DPP-4 Inhibitors offer safety and tolerability but modest efficacy.

Individual patient factors should guide selection, including cost, patient preference, comorbidities, and weight goals.

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